

NISHANTH ANANTHA

Staff Technical Program Manager · AI Systems Builder · Portfolio Governance · Edge AI
nishanth.anantha87@gmail.com · +1 (213) 327-7697 · San Marcos, CA · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

ABOUT

Staff TPM with 14+ years owning cross-functional portfolio visibility and delivery accountability across hardware, embedded SW, and AI systems programs. Builder, Connector, Translator: I build production AI systems (Claude API + MCP + ensemble ML), connect HW/FW/SW/AI teams around a shared execution reality, and translate business commitments into engineering capacity plans — surfacing tradeoffs before they become misses. Technical depth spans FPGA/embedded systems, SoC integration, and production agentic AI — enough to engage credibly with hardware, embedded, cloud, and computer vision teams without domain specialists present.

WHAT I'VE DELIVERED

- Coordinated release governance across 12 engineering teams, managing 5 product lines simultaneously — single source of truth for commitments, capacity, dependencies, and status; surfaced cross-team blockers to Director and VP before they became schedule misses; 29%→94% release success rate across 30+ program releases in 18 months
- Built production agentic AI portfolio intelligence system (Claude API + MCP + GitHub + JIRA + Smartsheet + ensemble ML) — autonomous signal collection across hardware and SW programs with SHAP explainability; adopted by Director and VP stakeholders; 40% release risk reduction
- Translated business commitments into engineering capacity plans — chaired cross-ocean Change Control Board coordinating HW/FW/SW programs across US, Europe, and Asia; managed prioritization decisions when capacity and demand conflicted
- Managed \$2M+ validation infrastructure — vendor qualification, procurement, and platform delivery with 70% field defect reduction; \$3M+ customer savings via rapid cross-functional escalation resolution
- Built NIM Release Risk Agent (NVIDIA NeMo Agent Toolkit + NIM-hosted Llama 3.1 70B) — a deliberate NVIDIA-native counterpart to the production Claude+MCP system, with Phoenix tracing and a custom evaluation harness

EXPERIENCE

Staff Technical Program Manager — Portfolio Intelligence & NPI Governance *Jan 2024 – Present*

ASML · San Diego, CA · Team: 20

- Owned company-level program portfolio — maintained single source of truth for commitments, capacity, dependencies, and status across 12 engineering teams and 5 product lines; designed executive reporting cadences giving VP leadership clear visibility without noise
- Built and deployed production agentic AI portfolio intelligence system (Claude API + MCP + GitHub REST + JIRA + Smartsheet + PyTorch + GBM + DistilBERT + SHAP + FastAPI) — security hardened; feeds Go/NoGo operating reviews adopted by Director and VP
- Served as primary interface between executive leadership and engineering — translated business commitments into engineering capacity plans; surfaced tradeoffs before they became misses; drove prioritization when capacity and demand conflicted

Sr. Technical Program Manager — SW/HW Integration & Platform Programs *Mar 2019 – Dec 2023*

ASML · San Diego, CA · Team: 10

- Built AI program operating model from scratch — execution forums, Change Control Board, release cadences, dependency tracking, and portfolio health reporting; introduced governance in a high-growth engineering environment
- Chaired cross-ocean CCB — coordinated HW/FW/SW change velocity across US, Europe, and Asia; translated engineering change impact into business tradeoffs for senior leadership weekly
- Drove CI/CD transformation org-wide (Jenkins, GitHub Actions, Docker) — 65% faster delivery; built ensemble ML risk predictor (PyTorch + GBM + DistilBERT, 13 signals, SHAP); 35% reduction in last-minute integration failures
- Reduced customer-facing defects 50%; resolved \$3M+ in customer escalations; managed \$2M+ validation infrastructure with 70% field defect reduction

Sr. Software Engineer / Team Lead / Scrum Master *Nov 2016 – Mar 2019*

ASML · San Diego, CA · Team: 7

- Engineer-to-TPM arc: shipped Python automation to production while owning SAgile planning, backlog management, and release governance
- Lifted release success from 29% to 94% in 18 months; resolved \$3M+ customer escalation; awarded for Leadership and Ownership

SWI Consultant — NPI Bring-Up *Jan 2021 – Jun 2021*

HMI (acq. by ASML) · San Jose, CA

- Built entire execution framework from zero in a high-growth startup — portfolio baseline, dependency map, Agile cadence, validation coverage 50%→80%; delivered major customer commitment one month early

Sr. Electrical Engineer / SW Engineer — Embedded & Edge Systems 2009 – 2016

Caterpillar · Volvo Group · Abbott

- Caterpillar/ARI: Embedded C on dSPACE ATEs (Freescale MPC535/SoC); HIL testbenches; Lauterbach firmware debugging; Python automation; \$500K+ field escalation resolution
- Volvo Group: FPGA-based cRIO LabVIEW RT ATE simulating vehicle powertrain environments — CAN J1939/J1587, NI TestStand; global owner of Bluetooth infotainment SW test platform; Patent Idea of the Year 2012 — Volvo Global Patent Council
- Abbott: LabVIEW CVI ATE testbench for heart pacer IC/wafer characterization; GPIB/VISA instrument drivers; FDA-compliant documentation

AI SYSTEMS & FEATURED BUILDS

Agentic Portfolio Intelligence System

Claude API · MCP · GitHub REST · JIRA · Smartsheet · PyTorch · GBM · DistilBERT · SHAP · FastAPI

- Agentic Claude loop routes tool calls across live operational data sources — synthesizing 13 execution signals into portfolio risk score with SHAP factor attribution; security hardened; feeds executive operating reviews; adopted in production

NIM Release Risk Agent

NVIDIA NeMo Agent Toolkit (NAT) · NIM (Llama 3.1 70B) · Phoenix · Ragas

- Built a NVIDIA-native ReAct agent (NAT + NIM-hosted Llama 3.1 70B) scoring SW release risk from live GitHub signals via a production ensemble ML API — a deliberate NVIDIA-stack counterpart to the production Claude+MCP system, built for hands-on depth with NAT and NIM
- Used Phoenix tracing to identify and merge a redundant tool-call pattern, cutting end-to-end latency ~20% (66s → 51s), verified against traced spans
- Built a custom NAT evaluator after Ragas's RAG-oriented metric couldn't score a tool-calling agent — 0.8/1.0 grounded-correctness, with the gap traced to a known LLM-judge limitation

Edge CV Inference System — AI Builder Project

YOLOv3 · cvlib · FastAPI · Docker · TensorFlow · Python

- YOLOv3 person/vehicle detector deployed as a versioned FastAPI service with structured telemetry, p50/p95 latency tracking, and person/vehicle/street object filtering
- Benchmarked cold-start penalty (13.9x: 1073ms → 77ms cached) and confidence threshold operating point (0.3 validated); scoped multi-stream architecture constraints and a documented production path to NVIDIA DeepStream / Jetson NPU

RAG Conversational AI Agent with Voice Evaluation Pipeline

HuggingFace · ChromaDB · W&B LLM Evaluation · Deepgram STT/TTS · Streaming Inference

- End-to-end RAG system with voice pipeline; offline eval (BLEU/ROUGE, W&B LLM Evaluation) → online results of 82% CSAT, 65% deflection, 1.1s p50 latency; multi-modal evaluation with Deepgram STT/TTS

SKILLS & TOOLS

Portfolio Governance: Cross-functional portfolio management, capacity planning, demand management, executive reporting, operating rhythm design, single source of truth systems, dependency management, PMI-PMP, Agile/SAFe

Technical Fluency: Hardware NPI, embedded SW/HW (FPGA, cRIO, LabVIEW RT, CAN), SoC platforms, AI/ML systems, CI/CD, edge-deployed systems, real-time systems

AI / ML Stack: Claude API (Anthropic certified), Agentic AI (DeepLearning.AI), MCP, LangGraph, NVIDIA NeMo Agent Toolkit (NVIDIA certified), PyTorch, TensorFlow, HuggingFace, RAG, ChromaDB, SHAP, Google Vertex AI

Tools & Platforms: JIRA, Smartsheet, Confluence, MS Project, Python, SQL, GitHub Actions, Jenkins, Docker, FastAPI, REST, AWS Certified, Power BI

EDUCATION

M.S. Electrical & Electronics Engineering — University of Southern California (USC), 2014

B.E. Electronics & Communication Engineering — SJCE / VTU, India, 2009

CERTIFICATIONS

PMI-PMP (Active) · Building with Claude API — Anthropic (Apr 2026) · NVIDIA NeMo Agent Toolkit — NVIDIA (Jul 2026) · Agentic AI — DeepLearning.AI (Jul 2026) · Machine Learning in Production (MLOps) — DeepLearning.AI (Jun 2026) · Deep Learning with TensorFlow — Kaggle (Apr 2026) · AWS Certified Cloud Practitioner (2024)